Simultaneous endpoint saturation and the positive inertia of Chern periods

23 September 2026. Baseline: `8113dd1`. This note deduces a global constraint on actual Chern classes from the bounded-form endpoint theorem. The conclusion concerns the positive part of the intersection form. No equal-square, nonnegative-square, topological-square, or negative-square budget is imposed on the classes.

1. Statement from actual geometric inputs

Let `(M,g)` be a closed connected oriented smooth four-manifold, with the Einstein target normalized to `Ric = -g`. Let `a_1,...,a_m`, `m >= 2`, be real first Chern classes of specified Spin^c structures whose unperturbed Seiberg–Witten equations are solvable for every metric conformal to `g`. Each structure is held fixed during conformal variation. The classes need not be linearly independent or have equal squares. Put

$$V = \mathrm{Vol}(g), \quad p_i = (a_{i,g}^+)^2 > 0, \quad r = \fint |\mathrm{Ric}+g|^2, \quad w = \sqrt{6}|W^+|,$$
$$\delta_i = \frac{3}{2}\int (s^2 + 2w^2) - 72\pi^2 p_i \geqslant 0, \qquad \kappa = \max_i\{r + \delta_i/V\} \leqslant 1/2880. \tag{1}$$

Let `lambda_1>0` be the first positive scalar-Laplacian eigenvalue and set

$$G(\kappa) = 13\kappa + 4\sqrt{2\kappa}, \qquad J = \left(300\sqrt{\kappa} + 12\sqrt{G(\kappa)/\lambda_1}\right)^2. \tag{2}$$

For `kappa=0`, set `J=0`.

Theorem (signed periods and positive inertia). There are constant signs `s_1=1,s_2,...,s_m` such that, for every `i`,

$$\boxed{\frac{2\pi^2}{V}\left((a_i - s_i a_1)_g^+\right)^2 \leqslant J.} \tag{3}$$

Consequently, for every real vector `t` with `sum_i t_i s_i=0`,

$$\boxed{\frac{2\pi^2}{V}\left(\sum_i t_i a_i\right)^2 \leqslant J\left(\sum_{i=2}^m |t_i|\right)^2 \leqslant (m-1)J\sum_i t_i^2.} \tag{4}$$

The first square in (3) is the positive Hodge norm; the square in (4) is the full, indefinite cup-product square.

The following matrix form records more of the structure. Define the normalized cup and positive-period matrices

$$Q_{ij} = \frac{2\pi^2}{V} a_i \cdot a_j, \qquad P_{ij} = \frac{2\pi^2}{V} a_{i,g}^+ \cdot a_{j,g}^+, \qquad q_1 = P_{11}.$$

There are positive semidefinite matrices `R,N` and a real vector `alpha` such that

$$\boxed{P = q_1 \alpha\alpha^{\mathsf{T}} + R, \quad Q = q_1 \alpha\alpha^{\mathsf{T}} + R - N, \quad \operatorname{tr} R \leqslant (m-1)J,} \tag{5}$$

$$\alpha_1 = 1, \qquad |\alpha_i - s_i| \leqslant \sqrt{J/q_1}, \qquad \frac{1}{2} \leqslant q_1 \leqslant \frac{3}{2}. \tag{6}$$

The positive-period matrix itself has the sharper entrywise estimate

$$\boxed{\max_{i,j} |P_{ij} - q_1 s_i s_j| \leqslant \kappa/36 + 2J.} \tag{6a}$$

This uses the exact common-functional relation among the class lengths, not merely Cauchy–Schwarz applied to the `L^2` approximation. In particular, if `nu_2(Q)` denotes the second largest eigenvalue,

$$\boxed{\nu_2(Q) \leqslant (m-1)J.} \tag{7}$$

No smallness of `J` is needed for these finite inequalities. They become restrictive when `kappa` and `G(kappa)/lambda_1` are small. In particular, `lambda_1 >> sqrt(kappa)` is sufficient along a sequence; a fixed positive lower bound for `lambda_1` is stronger than necessary.

2. Two useful cup-product obstructions

For two classes `a,c`, (3) immediately gives

$$\boxed{\frac{2\pi^2}{V}\left(a^2 + c^2 - 2|a \cdot c|\right) \leqslant J.} \tag{8}$$

Thus positive cup square for both `a-c` and `a+c` obstructs simultaneous saturation when the scalar gap is large relative to the error. This criterion does not require either class individually to have positive square. It may be vacuous, which is correctly reported by a nonpositive left side.

More generally, if `nu_2(Q)>=theta>0` and

$$\kappa \leqslant \frac{\theta}{360000(m-1)},$$

then (7) gives the explicit alternative

$$\boxed{\lambda_1 \leqslant \frac{576(m-1)G(\kappa)}{\theta}.} \tag{9}$$

Indeed, (7) implies `sqrt(theta/(m-1)) <= 300 sqrt(kappa)+12 sqrt(G/lambda_1)`; the first summand is at most half the left side. If `kappa=0`, a positive `theta` is impossible. These statements obstruct a proposed metric and its specified actual classes; they do not rule out all Einstein metrics on an arbitrary smooth manifold.

For fixed classes and bounded volume, a positive second eigenvalue of their unnormalized cup matrix therefore excludes a sequence with `kappa -> 0` and `G(kappa)/lambda_1 -> 0`. If volume grows, the necessary normalization in (7) must be retained: positive inertia alone, without a quantitative positive eigenvalue divided by volume, is insufficient.

3. An elementary phase lemma with a squared error

All expectations below use normalized volume. Let `u,v` be smooth sections of the same Euclidean rank-three bundle, with `|u|,|v|<=sqrt(6)`. Write

$$a = |u|^2, \quad b = |v|^2, \quad h = \langle u, v\rangle, \quad \mathscr{Q}(u) = u \otimes u - \frac{|u|^2}{3} I, \quad \sigma = \operatorname{sign} h \in \{-1, 1\},$$

resolving `h=0` by `sigma=1`. Pointwise,

$$|\mathscr{Q}(u) - \mathscr{Q}(v)|^2 = \frac{2}{3}(a-b)^2 + 2(ab - h^2). \tag{10}$$

Two useful estimates, including at zeros, are

$$|u - \sigma v|^2 \leqslant 2\Big((a-1)^2 + (b-1)^2\Big) + 2|\mathscr{Q}(u) - \mathscr{Q}(v)|^2, \tag{11}$$

$$|h - \sigma|^2 \leqslant 2\Big((a-1)^2 + (b-1)^2\Big) + |\mathscr{Q}(u) - \mathscr{Q}(v)|^2. \tag{12}$$

For (11), if `a+b>=1`, use

$$|u - \sigma v|^2 = \frac{(a-b)^2 + 4(ab - h^2)}{a + b + 2|h|} \leqslant 2|\mathscr{Q}(u) - \mathscr{Q}(v)|^2.$$

If `a+b<1`, the left side is at most `a+b`, whereas `2((a-1)^2+(b-1)^2)>=(a+b-2)^2>=a+b`. For (12), let `z=sqrt(ab)` and, when `z>0`, `t=|h|/z`. Then `0<=t<=1` and

$$(zt-1)^2 \leqslant 2(z-1)^2 + 2z^2(1-t)^2, \quad (z-1)^2 \leqslant (a-1)^2 + (b-1)^2, \quad 2z^2(1-t)^2 \leqslant 2(ab - h^2).$$

The middle inequality follows because `sqrt(ab)` lies between `a` and `b`; the zero case follows directly. Equation (10) completes the proof.

Suppose `E|h-sigma|^2<=T` and `E|dh|^2<=D`. Poincare and the triangle inequality for orthogonal projection off the constants give

$$\sqrt{\operatorname{Var}\sigma} \leqslant \sqrt{\operatorname{Var} h} + \|h - \sigma\|_2 \leqslant \sqrt{D/\lambda_1} + \sqrt{T}. \tag{13}$$

There is a constant sign `s` with

$$\mathbb{E}|\sigma - s|^2 = 2(1 - |\mathbb{E}\sigma|) \leqslant 2 \operatorname{Var}\sigma. \tag{14}$$

If `E|u-sigma v|^2<=F`, it follows that

$$\|u - sv\|_2 \leqslant \sqrt{F} + \sqrt{12}\Big(\sqrt{D/\lambda_1} + \sqrt{T}\Big). \tag{15}$$

All norms here are normalized `L^2` norms. No derivative of the measurable sign, unit eigenvector, or normalized form is taken. The argument only needs the Poincare estimate for these finitely many smooth scalar functions `h`; a gap on the whole scalar spectrum is a convenient sufficient condition.

4. Applying the lemma to the actual endpoint forms

Let `H_i` represent `a_{i,g}^+` harmonically and put `xi_i=-sqrt(2) pi H_i`. The actual bounded-form theorem in the endpoint application, equations (2)–(4), gives smooth `u_i=omega_i/sqrt(2)` with arbitrarily small simultaneous smoothing allowances. Suppressing those allowances until the limiting step, it gives

$$\begin{aligned} |u_i| &\leqslant \sqrt{6}, \qquad \mathbb{E}|u_i-\xi_i|^2 \leqslant 49\kappa/2, \\ \mathbb{E}(|u_i|^2-1)^2 &\leqslant 180\kappa, \qquad \mathbb{E}|W^+ + \mathscr{E}(u_i)|^2 \leqslant 720\kappa, \\ \mathbb{E}|\nabla u_i|^2 &\leqslant G(\kappa)/2. \end{aligned} \tag{16}$$

The same Weyl tensor occurs for every class, so

$$\mathbb{E}|\mathscr{E}(u_i)-\mathscr{E}(u_1)|^2 \leqslant 2880\kappa. \tag{17}$$

Equations (11)–(12), and the differential estimate `|d<u_i,u_1>|^2 <=12(|nabla u_i|^2+|nabla u_1|^2)`, give

$$F = 6480\kappa, \qquad T = 3600\kappa, \qquad D = 12G(\kappa). \tag{18}$$

Thus (15), followed by the two harmonic-alignment errors, gives a constant sign `s_i` with

$$\|\xi_i - s_i\xi_1\|_2 \leqslant \left(7\sqrt{2}+36\sqrt{5}+120\sqrt{3}\right)\sqrt{\kappa} + 12\sqrt{G(\kappa)/\lambda_1} \leqslant 300\sqrt{\kappa} + 12\sqrt{G(\kappa)/\lambda_1}. \tag{19}$$

The coefficient estimate is elementary: use `sqrt(2)<3/2`, `sqrt(5)<9/4`, and `sqrt(3)<26/15`, giving an upper bound `299.5`.

To restore the allowances rigorously, choose a sequence of positive allowances tending to zero, obtaining one finite sign vector for each choice. Pass to a subsequence with constant sign vector. The harmonic forms, their norms, and the scalar eigenvalue are fixed. All the errors in (16)–(19) tend to their displayed bounds. No convergence of spinors, connections, or measurable sign functions is asserted. This proves (3). Since a full cup square is at most its positive Hodge square, summing the signed differences proves (4).

For completeness, (16) implies

$$\left(\sqrt{1-\sqrt{180\kappa}} - 7\sqrt{\kappa/2}\right)^2 \leqslant q_1 \leqslant \left(\sqrt{1+\sqrt{180\kappa}} + 7\sqrt{\kappa/2}\right)^2. \tag{20}$$

The lower parenthesis is positive on `kappa<=1/2880`, and this interval gives `1/2<=q_1<=3/2`.

5. The matrix decomposition and its precise limitation

Use the Hilbert space of self-dual forms with normalized `L^2` inner product. Orthogonally split

$$\xi_i = \alpha_i\xi_1 + \eta_i, \qquad \alpha_i = \frac{\langle \xi_i, \xi_1\rangle_{L^2(d\mu/V)}}{q_1}, \quad \langle \eta_i, \xi_1\rangle = 0.$$

Here alpha_1=1, eta_1=0, and (3) gives ||eta_i||_2^2<=J and |alpha_i-s_i|<=sqrt(J/q_1). Set R_ij=<eta_i,eta_j> and let N be the normalized positive Gram matrix of the anti-self-dual harmonic representatives. Then R,N>=0, tr R<= (m-1)J, and Hodge decomposition gives (5) exactly. On the codimension-one subspace perpendicular to alpha, the quadratic form of Q is at most ||R||_op<=tr R. The min–max principle proves (7).

To prove (6a), put q_i=P_ii. Subtracting the definitions of the deficits in (1) gives the exact identity

$$q_i - q_1 = -\frac{\delta_i - \delta_1}{36V}, \qquad |q_i - q_1| \leqslant \kappa/36. \tag{21}$$

The signed forms s_i xi_i are each within sqrt(J) of xi_1, so the squared distance between any two is at most 4J. Polarization therefore gives

$$q_1 - \kappa/36 - 2J \leqslant s_i s_j P_{ij} \leqslant q_1 + \kappa/36,$$

proving (6a). No extra square root of J is lost in the positive Gram matrix estimate.

It would be incorrect to delete N or replace (5) by closeness of Q to a rank-one sign matrix. The bounded-form estimates control the positive projections. For example, in an abstract intersection space of signature (1,m), choose an orthonormal positive vector e_0 and negative vectors e_i, and let a_i=e_0+L e_i. Their positive Gram matrix is exactly 11^T, while their full Gram matrix is 11^T-L^2 I. This illustrates the missing negative information even when positive-period alignment is exact. It is an algebraic test of the implication, **not** an assertion that these classes are actual monopoles on a metric saturating (1). Classical gauge equality rigidity can impose further restrictions at exact saturation.

Similarly, distinct Spin^c labels, torsion differences, or the pair a,-a do not by themselves supply a positive period obstruction. An integral lattice alone gives no lower bound on the positive Hodge norm of all nonzero vectors; one must use a verified positive cup square, a quantitative positive eigenvalue, or additional period data.

6. Exact increment over the earlier arguments

Round 18, Sections 1–4, assumes common positive square a_i^2=A>=2chi+3tau and uses scalar-energy saturation to control the **full** determinant curvatures. It obtains an intersection sign-correlation polytope and low scalar eigenvalues from covariance. Those remain stronger in their particular regime and are not new here. Round 2, Sections 3–4, already handles measurable line alignment and Walsh phase separation at a different calibration.

The added result is (3)–(7) for the actual all-scalar endpoint input, with arbitrary individual squares and no negative-square budget. It extracts a positive-inertia obstruction from harmonic Chern periods, while preserving the uncontrolled negative Gram matrix. The squared phase estimates (11)–(14) yield an O(kappa+G(kappa)/lambda_1) error, rather than losing a square root simply by estimating the mean of the inner product. The common-functional identity (21) preserves that order in the positive Gram matrix. No growing low-mode cluster theorem is being claimed.

The classical scalar–Weyl curvature bound and its equality geometry come from C. LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Theorem 2.3 and Proposition 4.1. The local endpoint application provides the quantitative all-scalar harmonic alignment used here.

`line1_period_inertia_check.py` checks the exact tensor identities, phase inequalities' algebra, constants, and a finite model of (5). It does not certify gauge solvability or replace Poincare, Hodge theory, the smoothing argument, or the analytic endpoint theorem.